Vaastav Anand

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Education

PhD, Computer Science 2020-2026
Saarland University
Max Planck Institute for Software-Systems (MPI-SWS), Saarbruecken, Germany
Thesis: Building Efficient Modular Cloud Systems
Advisors: Antoine Kaufmann, Deepak Garg, (past: Jonathan Mace)

MSc, Computer Science 2018-2020
University of British Colombia, Vancouver, Canada
Thesis: Dara the explorer: coverage based exploration for model checking of distributed systems in Go.
Advisor: Ivan Beschastnikh, (2nd Reader: Margo Seltzer)

BSc, Computer Science 2013-2018
University of British Colombia, Vancouver, Canada

Employment

Postdoctoral Researcher, ETH Zurich 2026-current
EASL/Systems Group, Advisor: Dr. Ana Klimovic

Research Intern, Microsoft 2024
Azure Research - Systems

- Designed and Developed Cerulean, a tool for automatically generating microservice systems from natural language descriptions.
- Mentors: Rodrigo Fonseca, Alok Gautam Kumbhare
- Collaborators: Pedro Las-Casas, Celine Irvene, Jonathan Mace, Gagan Somashekar, Chetan Bansal, Ricardo Bianchini

Research Intern, MPI-SWS 2019

- Instrumented DeathStarBench applications with XTrace tracing for large-scale experiments for Sifter.

Undergraduate Research Assistant 2018
University of British Columbia, Under Ivan Beschastnikh

- Designed and developed Dara, a tool for model checking distributed systems.

Software Engineering Intern, NVIDIA 2017
MODS (Modular Diagnostics) Team

- Implemented memory repair sequences for faulty High Bandwith Memory(HBM).
- Designed, developed, and implemented a CUDA based full memory stress test for identifying faulty HBM cells.

Software Engineering Intern, NVIDIA 2016
MODS (Modular Diagnostics) Team

- Implemented a synchronization option for CUDA based linpack tests to synchronize CUDA kernel launches within 30μs across multiple GPUs.

Software Developer Intern, Thinkbox Software 2015-2016
Sequoia Team

- Designed, developed, and implemented the 3D PDF export feature for point cloud meshes in Sequoia.

Publications

Peer-Reviewed Publications

1. Iridescent: A Framework Enabling Online System Implementation Specialization. **Vaastav Anand**, Deepak Garg, Antoine Kaufmann. Under Review.
2. Enabling Multi-Dimensional Distributed Trace Comparison with Contrast. **Vaastav Anand**, Rodrigo Fonseca, Jonathan Mace, Antoine Kaufmann. Under Review.
3. Using Epistemic Observability in Agentic Systems for Combating Hallucinations. Tony Mason, **Vaastav Anand**, Practical Adoption Challenges of ML for Systems (co-located with SOSP'26), **PACMI'26**.
4. Peeking Under the Hood of Multi-Agent Systems. Tie Ma*, Yixi Chen*, **Vaastav Anand**, Alessandro Cornacchia, Amandio R. Faustino, Guanheng Liu, Shan Zhang, Hongbin Luo, Suhaib A. Fahmy, Zafar A. Qazi, Marco Canini. 1st ACM Conference on AI and Agentic Systems, System Demonstrations Track, **CAIS'26 Demos**.
5. Intent-Based System Design and Operation. **Vaastav Anand**, Yichen Li, Alok Gautam Kumbhare, Celine Irvene, Chetan Bansal, Gagan Somashekar, Jonathan Mace, Pedro Las-Casas, Ricardo Bianchini, Rodrigo Fonseca. Practical Adoption Challenges of ML for Systems (co-located with SOSP'25), **PACMI'25**.
6. DMAS-Forge: A Framework for Transparent Deployment of AI Applications as Distributed Systems. Alessandro Cornacchia, **Vaastav Anand**, Muhammad Bilal, Zafar Ayyub Qazi, Marco Canini. 1st Workshop on Systems for Agentic AI (co-located with SOSP'25), **SAA'25**.
7. Generating representative macrobenchmark microservice systems from distributed traces with Palette. **Vaastav Anand**, Matheus Stolet, Jonathan Mace, Antoine Kaufmann. 16th ACM SIGOPS Asia-Pacific Workshop on Systems (co-located with SOSP'25), **ApSys 2025**.
8. Automated Service Design with Cerulean (Project Showcase). **Vaastav Anand**, Alok Kumbhare, Celine Irvene, Chetan Bansal, Gagan Somashekar, Jonathan Mace, Pedro Las-Casas, Rodrigo Fonseca. 6th International Workshop on Cloud Intelligence/AIOps Workshop (co-located with ICSE), **AIOps 2025**.
9. Towards Using LLMs for Distributed Trace Comparison (Abstract). **Vaastav Anand**, Pedro Las-Casas, Rodrigo Fonseca, Antoine Kaufmann. 6th International Workshop on Cloud Intelligence/AIOps Workshop (co-located with ICSE), **AIOps 2025**.
10. Blueprint: A Toolchain for Highly Reconfigurable Microservice Applications. **Vaastav Anand**, Deepak Garg, Antoine Kaufmann, Jonathan Mace. In Symposium on Operating Systems Principles, **SOSP 2023**.
11. The Benefit of Hindsight: Tracing Edge Cases in Distributed Systems. Lei Zhang, Zhiqiang Xie, **Vaastav Anand**, Ymir Vigfusson, Jonathan Mace. In Networked Systems Design and Implementation, **NSDI 2023**.
12. The Odd One Out: Energy is not like Other Metrics. **Vaastav Anand**, Zhiqiang Xie, Matheus Stolet, Roberta De Viti, Thomas Davidson, Reyhaneh Karimipour, Safya Alzayat, Jonathan Mace. In **HotCarbon 2022**.
13. Smooth Kronecker: Solving the Combing Problem in Kronecker Graphs. **Vaastav Anand***, Puneet Mehrotra*, Daniel Margo*, Margo Seltzer. In Joint Workshop on Graph Data Management Experiences and Systems and Network Data Analytics, **GRADES-NDA 2020**.

14. Sifter: Scalable Sampling for Distributed Traces, without Feature Engineering. Pedro Las-Casas, Giorgi Papakerashvili, **Vaastav Anand**, Jonathan Mace. In Symposium on Cloud Computing, **SoCC 2019**.

Pre-Prints

1. Epistemic Observability in Predictive Systems: An Impossibility Result. Tony Mason, **Vaastav Anand**.
2. MAESTRO: Multi-Agent Evaluation Suite for Testing, Reliability, and Observability. Tie Ma*, Yixi Chen*, **Vaastav Anand**, Alessandro Cornacchia, Amandio R. Faustino, Guanheng Liu, Shan Zhang, Hongbin Luo, Suhaib A. Fahmy, Zafar A. Qazi, Marco Canini. In **arXiv 2026** `https://arxiv.org/pdf/2601.00481`.
3. SoK: Demystifying the multiverse of MPC protocols. Roberta De Viti, **Vaastav Anand**, Pierfrancesco Ingo, Deepak Garg. In **arXiv 2025** `https://arxiv.org/pdf/2512.11699`.
4. Chatting with Logs: An exploratory study on Finetuning LLMs for LogQL. Vishwanath Seshagiri, Siddharth Balyan, **Vaastav Anand**, Kaustubh Dhole, Ishan Sharma, Avani Wildani, José Cambronero, Andreas Züfle. In **arXiv 2025**, `https://arxiv.org/pdf/2412.03612`.
5. Columbo: Low Level End-to-End System Traces through Modular Full-System Simulation. Jakob Görgen, **Vaastav Anand**, Hejing Li, Jialin Li, and Antoine Kaufmann. In **arXiv 2024**, `https://arxiv.org/pdf/2408.05251`.
6. SoK: The Faults in our Graph Benchmarks. Puneet Mehrotra*, **Vaastav Anand***, Daniel Margo, Milad Rezaei Hajidehi, and Margo Seltzer. In **arXiv 2024**, `https://arxiv.org/pdf/2404.00766`.
7. Aggregate-driven trace visualizations for performance debugging. **Vaastav Anand**, Matheus Stolet, Thomas Davidson, Ivan Beschastnikh, Tamara Munzner, and Jonathan Mace. In **arXiv 2020**, `https://arxiv.org/pdf/2010.13681`.

Datasets

1. X-Trace distributed trace dataset for DeathStarBench. **Vaastav Anand** and Jonathan Mace. `https://gitlab.mpi-sws.org/cld/trace-datasets/deathstarbench_traces`
2. Fantasy Premier League Gameweek-By-Gameweek Dataset. Vaastav Anand. `https://github.com/vaastav/Fantasy-Premier-League`

Teaching

Instructor, Saarland University — 2025

- Reliability in Modern Cloud Systems — 2025

Teaching Assistant, Saarland University — 2021

- 1× TA for Distributed Systems (Core Course) — 2021

Graduate Teaching Assistant — 2018-2020
University of British Coluimbia, Department of Computer Science

- 1× TA for Computer Hardware and Operating Systems (CPSC 313) — 2020
- 1× TA for Graduate Operating Systems (CPSC 508) — 2019
- 1× TA for Distributed Systems (CPSC 416) — 2018

Academic Assistant 2018
Vancouver Summer Program
- Teaching Assistant for the Algorithms and the World Wide Web course.

Undergraduate Teaching Assistant 2014-2018
University of British Columbia, Department of Computer Science
- 1× TA for Introduction to Software Engineering (CPSC 210) 2018
- 1× TA for Advanced Operating Systems (CPSC 415) 2017
- 1× TA for Intermediate Algorithm Design and Analysis (CPSC 320) 2017
- 1× TA for Computer Hardware and Operating Systems (CPSC 313) 2016
- 1× TA for Introduction to Computer Systems (CPSC 213) 2015
- 3× TA for Models of Computation (CPSC 121) 2014-2015

Awards

Busy Beaver Award, Best Seminar in CS, Saarland University 2026
3rd Place, Graduate Category, ACM SRC Grand Finals 2025
1st Place, Graduate Catgeory, SOSP'24 SRC 2024
SoCC Student Scholarship 2019
2nd Place, Undergraduate Category, FSE'18 SRC 2018
SIGSOFT CAPS Award 2018
UBC International Tuition Award 2018-2019
Work Learn International Undergraduate Research Award 2018
Student Speaker, UBC Class of 2018 2018
UBC Faculty of Science, International Student Award 2015, 2018
ACM ICPC PacNW Regional Contest Division 2 Champion 2017
UBC Trek Excellence Scholarship 2016-17, 2017-2018
UBC Dean's Honor List 2014, 2015, 2017
UBC Computer Science Student Service Award 2015
GIIS Global Citizen Scholarship 2011-2013

Service

Academic Service
- Program Committee Member
  - SoCC 2026
  - EuroSys Posters 2025
  - CS-Can Student Symposium 2019
- Organizer
  - Tutorial on using Blueprint to accelerate Microservice Research @ SOSP'24
  - Panel on Reproducibility and Replication @ HotOS'23
  - Systems Trivia Co-Organizer @ HotOS'23, SOSP'21, HotOS'21
  - The Cornell, Maryland, Max Planck Pre-doctoral Research School (CMMRS), 2022
- Publicity Chair
  - Symposium on Operating Systems Principles (SOSP), 2027.
  - The Journal of Systems Research (JSys), 2022-2023.

Talks

Demystifying the multiverse of MPC protocols — Sep 2026
Event: SysteMPC Workshop 2026 (co-located with SOSP)
Organizers: Vasia Kalavri, John Liagouris, Mayank Varia

Blueprint: A Toolchain for Highly-Reconfigurable Microservices — Jun 2024
Event: CNI Seminar Series @ IISc Bangalore
Host: Sreeshma Shiv

Blueprint: A Toolchain for Highly-Reconfigurable Microservices — Jun 2024
Event: Azure Monitor Day of Learning @ Microsoft Redmond
Host: Kalyana Sundaram

Blueprint: A Toolchain for Highly-Reconfigurable Microservices — Aug 2023
Event: Trustworthy Systems Seminar @ UNSW Sydney
Host: Peter Chubb

Blueprint: A Toolchain for Highly-Reconfigurable Microservices — Aug 2023
Event: Basser Seminar Series @ University of Sydney
Host: Zhanna Sarsenbayeva

Millenial: Modular Microservice Macrobenchmarks — Jun 2021
Event: Tracing Jamboree @ Emory University (virtual)
Organizers: Avani Wildani, Ymir Vigfusson

Interests

Computing: Distributed Systems, Operating Systems, Software Engineering
Extra Curricular: Soccer, Linguistics, Cricket.